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Schrödinger's Navigator: Imagining an Ensemble of Futures for Zero-Shot Object Navigation

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Project Page: https://heyu322.github.io/Schrodinger-Navigator.github.io/

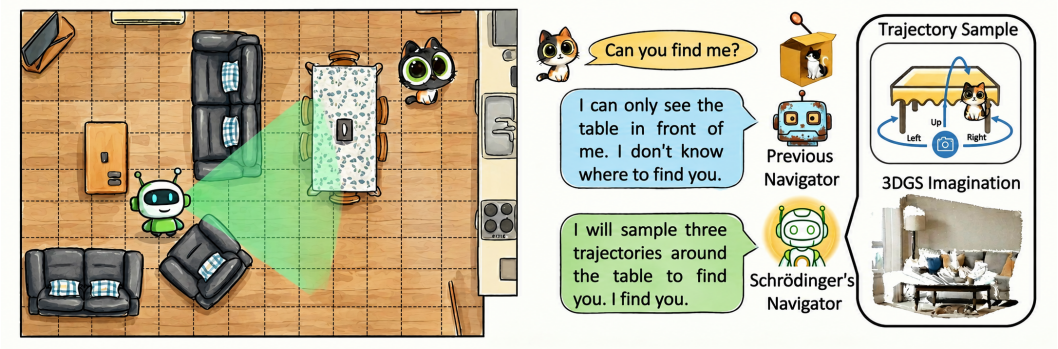


Figure 1: Real-world zero-shot object navigation often fails when the target (e.g., a cat) is hidden behind occlusions and surrounded by hazardous space. Conventional navigation systems perceive only the immediate occluder and are unable to infer what exists beyond it. Our Schrödinger's Navigator addresses this challenge by modeling the unobserved regions as plausible futures.

Abstract: Zero-shot object navigation (ZSON) requires robots to find target objects in unseen environments without task-specific fine-tuning or pre-built maps, a key capability for general-purpose service robots. Yet methods that perform well in simulation often degrade in cluttered real-world scenes with severe occlusion and latent hazards, where large unseen regions make single-scene inference brittle and unsafe. We propose **Schrödinger's Navigator**, a belief-aware framework that reasons at inference time over multiple *trajectory-conditioned imagined 3D futures*. Given candidate paths, a trajectory-conditioned 3D world model predicts hypothetical observations and maintains a superposition of plausible scene realizations rather than committing to one map. An adaptive occluder-aware sampler directs imagination to uncertainty-critical regions, while a *Future-Aware Value Map* (FAVM) aggregates imagined futures for robust, proactive action selection. Experiments in simulation and on a physical Go2 quadruped show that Schrödinger's Navigator outperforms strong ZSON baselines, improving hidden-target discovery and risk-aware waypoint selection in occlusion-heavy navigation scenarios. These results highlight imagined 3D futures as a promising foundation for a modular, scalable, and generalizable framework for zero-shot navigation in uncertain real-world environments.

Keywords: Zero-Shot Object Navigation, Generative Scene Imagination

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1 Introduction

Object navigation is a fundamental capability for mobile robots operating in human-centered environments [1, 2, 3]. Practical service robots must find requested objects in previously unseen scenes without pre-built maps [4, 5, 6] or costly scene-specific retraining [7, 8]. Zero-shot object navigation (ZSON) formalizes this setting by requiring a robot to locate a specified object in a novel environment without task-specific fine-tuning [9, 10].

Despite strong results in simulation, current ZSON systems often degrade in cluttered real-world scenes with severe occlusions and latent hazards [11, 10, 12, 13]. We define a latent hazard as an initially unobserved, unannotated region that may cause collision, unsafe traversal, or task failure if traversed [14, 15, 16]. In these scenes, targets and risks may only become visible from revealing viewpoints, as illustrated by the occluded cat in Figure 1. Yet many existing methods act on a single inferred scene interpretation, making them prone to missing hidden targets or choosing unsafe paths through uncertain regions [17, 18].

To address this limitation, we draw inspiration from Schrödinger's thought experiment on uncertainty and propose **Schrödinger's Navigator**, a principled navigation framework that treats unobserved space as a superposition of plausible future worlds and reasons over them before committing to an action, as illustrated in Figure 1. Rather than completing the scene once, the agent reasons at inference time over multiple *trajectory-conditioned imagined 3D futures*. This enables the robot to evaluate how the world may appear along different candidate paths and plan not only from what is observed, but also from what could be revealed behind occlusions.

At the core of our framework is a trajectory-conditioned 3D world model [19], which differs from prior world-model navigation methods [20, 21, 22] in how imagined futures are generated and used. Given egocentric visual observations and sampled trajectories, the model predicts hypothetical 3D observations along each path, maintaining multiple plausible realizations of unobserved space. To make imagination efficient and decision-relevant, we introduce an *adaptive, occluder-aware trajectory sampling* strategy that uses local geometry and uncertainty to focus predictions on regions where occlusion is most likely to affect navigation, avoiding costly dense rollouts over the full scene. The imagined 3D predictions are spatially aligned and fused into a *Future-Aware Value Map* (FAVM). Unlike value maps built only from observed states or short-horizon predictions, FAVM aggregates evidence across candidate futures and assigns value to currently unobserved regions according to their expected utility. The policy can reduce uncertainty, expose likely target locations, and avoid hazardous regions without requiring dense global reconstruction or extra exploratory detours.

We evaluate Schrödinger's Navigator in Habitat and on a physical Go2 quadruped across ObjectNav and occlusion-heavy scenarios. Results show that future-aware reasoning preserves competitive ObjectNav performance while improving behavior when targets or risks are initially hidden. By reasoning over imagined 3D futures at inference time, our method helps reveal hidden objects, avoid uncertain blind spots, and generalize to zero-shot navigation under real-world uncertainty.

In summary, our contributions are as follows:

- We identify a fundamental limitation of existing zero-shot object navigation methods under severe occlusion and latent hazards: their reliance on single-scene inference without explicit reasoning over uncertainty in unobserved space.
- We propose **Schrödinger's Navigator**, a belief-aware navigation framework that reasons over multiple *trajectory-conditioned imagined 3D futures*. The framework combines a trajectory-conditioned 3D world model, adaptive occluder-aware trajectory sampling, and a *Future-Aware Value Map* (FAVM) for proactive uncertainty-aware planning.
- We demonstrate in both Habitat simulation and real-world Go2 deployment that explicit reasoning over imagined 3D futures improves robustness in occlusion-heavy object search, earlier hidden-target discovery, and risk-aware waypoint selection.

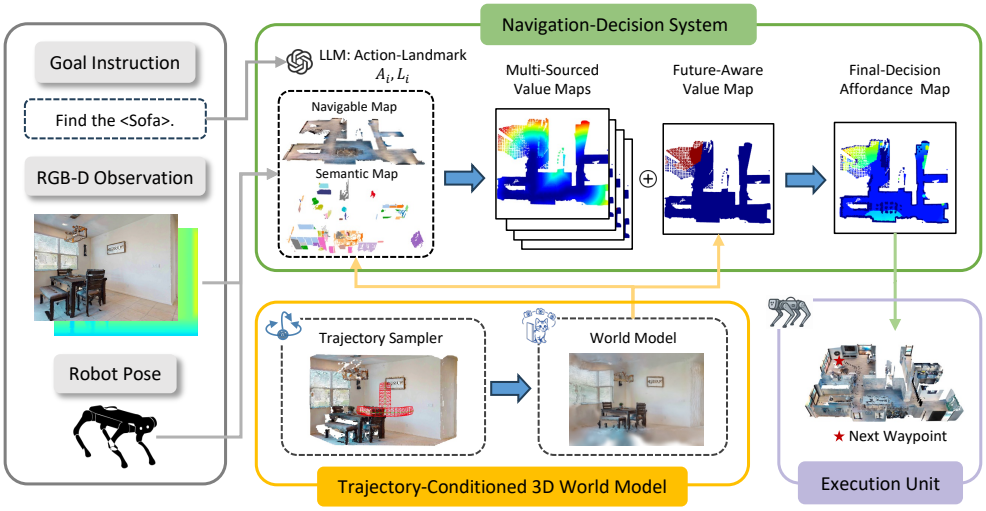


Figure 2: Overview of the Schrödinger's Navigator. Given the navigation context, the system adaptively samples candidate trajectories around occluded and uncertain regions. A trajectory-conditioned 3D world model then imagines future 3DGS observations along each path to expose hidden structures, target cues, and risks. These imagined futures are fused with current observations to construct a Future-Aware Value Map, aggregating trajectory-level utility and risk for robust waypoint selection. Finally, the selected navigation target is executed through the corresponding local controller.

2 Related Work

Object Navigation (ObjectNav) [23] studies embodied agents that locate target objects in unseen environments. *Task-trained* approaches based on reinforcement or imitation learning [8, 1, 24, 25] require large-scale training and often generalize poorly to cluttered real-world scenes, making sim-to-real deployment challenging. *Zero-shot* approaches instead leverage pretrained vision-language models [26, 9, 10] or large language models [27, 28, 29] for reasoning and planning without task-specific training [12, 10, 29, 30]. These methods improve semantics-driven exploration through multimodal target embeddings [9, 10], vision-language frontier maps [12], instruction-based prompting [31, 28], and adaptive fusion of semantic and geometric cues [32, 33, 34, 35, 36]. However, zero-shot ObjectNav remains brittle in realistic, occlusion-heavy environments and under unknown risks [10, 12], because value estimation is still dominated by visible frontiers and lacks explicit reasoning about hidden areas behind occluders, leading to myopic exploration.

Imagination for Navigation leverages generation models to simulate future observations and support planning [37, 20, 38, 21]. Related world-model methods learn predictive dynamics and roll out trajectories in latent space [39, 40]. Recent navigation world models further predict egocentric visual streams. NWM [20] predicts future trajectories and views for path ranking, while NavigateDiff [38] uses diffusion-based visual prediction as a zero-shot navigation assistant. Language-conditioned variants, such as VISTA and VISTAv2 [41, 42, 22], generate landmark images or instruction-conditioned future views and use their alignment to guide action selection. Other works imagine unseen scenes or predict occupancy maps to complete unobserved spaces for exploration [43, 44, 45, 46, 47]. These works are mainly studied under visual prediction, or vision-and-language navigation protocols. Unlike prior works, Schrödinger's Navigator imagines future observations along multiple paths with a trajectory-conditioned 3D world model and constructs a FAVM, enabling explicit reasoning about occlusions, risks, and blind-spot targets for robust zero-shot navigation.

3 Method

We introduce **Schrödinger's Navigator**, a zero-shot object navigation framework that reduces occlusion-induced uncertainty by reasoning over multiple trajectory-conditioned imagined 3D futures. At each decision step, the robot receives the current navigation context, including the goal instruction, RGB-D observation, robot pose, and accumulated map. Instead of committing to a single inferred scene, our method samples a compact set of occluder-aware candidate trajectories, imagines future 3D scenes along them, and aggregates the resulting cues into a future-aware affordance map for waypoint selection.

3.1 Trajectory-Conditioned 3D Future Imagination

Adaptive occluder-aware trajectory sampling. A single imagined path may miss targets or hazards hidden behind occlusions. To obtain complementary views with limited computation, we sample a small trajectory ensemble around the dominant occluder in the view. Specifically, we construct a local point cloud from the depth observation, remove ground points, and select the primary forward occluder according to projected occupancy, distance, and alignment with the robot's heading. Its boundary provides a local geometric anchor for trajectory generation.

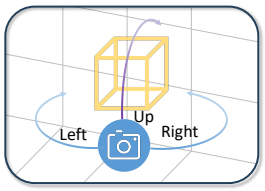


Figure 3: Occluder-aware camera trajectory sampling.

As shown in Figure 3, we instantiate three trajectory types $\mathcal{V} = \{L, R, U\}$, corresponding to left-bypass, right-bypass, and over-the-top views. For each type $v \in \mathcal{V}$, the sampler generates a camera trajectory $\tau_t^{(v)} = \mathcal{F}_{\text{samp}}(v; B_t, q_t, d_c, N_{\text{pose}})$, where B_t is the estimated occluder boundary, q_t is the current robot pose, d_c is a nominal orbit-distance parameter, and N_{pose} is the number of sampled camera poses. This compact trajectory set is designed to cover potentially visible regions on both sides of and above the occluder while avoiding dense sampling over the full action space. The three trajectories are generated from the same RGB-D observation to provide complementary virtual camera viewpoints, rather than independent world-state samples or robot-executed paths. Details of occluder boundary extraction, occluder selection, and trajectory parameterization are provided in Appendix B.

Trajectory-conditioned 3D world model. As shown in Figure 4, given each sampled trajectory $\tau_t^{(v)}$, we use a trajectory-conditioned 3D world model Φ_θ to generate a hypothetical future 3DGS:

$$\hat{\mathcal{G}}_t^{(v)} = \Phi_\theta \left(I_t, q_t, \tau_t^{(v)} \right),$$

where I_t denotes the current RGB observation. We adopt FlashWorld [19] as the backend because it can generate 3D-consistent scenes efficiently. Each generated scene represents a plausible realization of the currently unobserved region conditioned on one candidate path.

Since the generated 3DGS scene may not be metrically aligned with the robot's environment, we use the observed depth D_t to perform metric scale alignment before fusion. We first render a depth map $\hat{D}_t^{(v)}$ from the generated scene at the current viewpoint and estimate a global scale factor using valid pixels:

$$s_t^{(v)} = \underset{p \in \Omega_t^{(v)}}{\text{median}} \frac{D_t(p)}{\hat{D}_t^{(v)}(p)},$$

where $\Omega_t^{(v)}$ is the set of pixels with valid observed and rendered depths. The generated scene is then scaled and transformed into the world frame as $\tilde{\mathcal{G}}_t^{(v)} = \mathcal{A}(\hat{\mathcal{G}}_t^{(v)}, s_t^{(v)}, q_t)$, where $\mathcal{A}(\cdot)$ denotes scale alignment and coordinate transformation. The aligned scenes are fused into a unified future scene $\mathcal{G}_t^{\text{img}} = \bigcup_{v \in \mathcal{V}} \tilde{\mathcal{G}}_t^{(v)}$. We further transfer semantic labels from 2D segmentation to visible Gaussian primitives using depth-consistent projection and multi-view voting. Implementation details of 3DGS alignment, semantic label transfer, and scene fusion are provided in Appendix C.

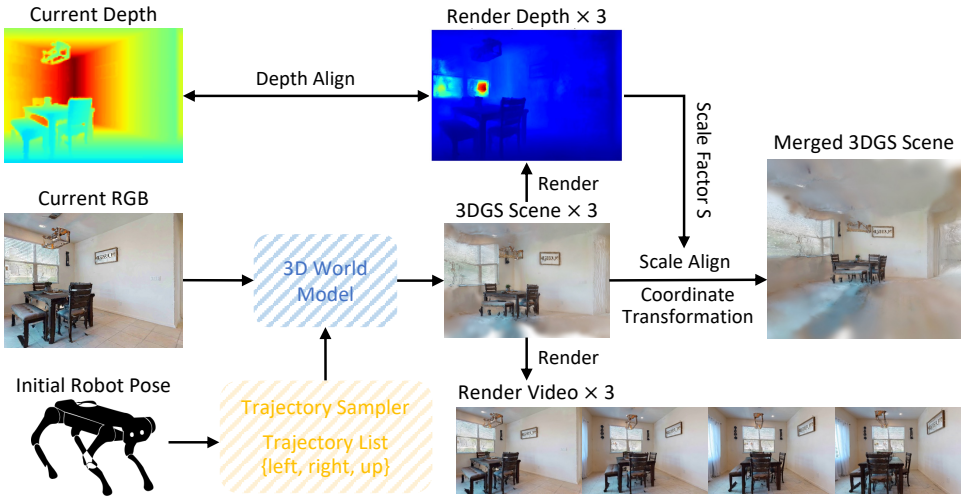


Figure 4: Overview of trajectory-conditioned 3D imagination: Given the current observation and robot pose, an occluder-aware sampler generates candidate trajectories that condition a 3D world model to predict future 3DGS scenes. These scenes are depth-aligned to recover a consistent scale, transformed into the world frame, and fused into a unified 3DGS scene for future-aware planning.

3.2 Future-Aware Navigation Decision

Future-Aware Value Map. The imagined scene $\mathcal{G}_t^{\text{img}}$ provides cues about free space, possible target locations, and risky regions beyond the current observation. We convert these cues into a *Future-Aware Value Map* (FAVM), denoted by m_t^{FA}, over candidate navigation positions $\mathcal{P}_t = \{\mu_g \mid g \in \mathcal{G}_{\text{nav}}\}$, where μ_g is the center of a navigable Gaussian:

$$m_t^{\text{FA}}(p) = \mathcal{N}(\alpha_{\text{sem}} S_t(p) + \alpha_{\text{exp}} E_t(p) - \alpha_{\text{risk}} R_t(p)), \qquad p \in \mathcal{P}_t,$$

where $S_t(p)$ measures semantic relevance to the target, $E_t(p)$ estimates the local coverage of newly imagined free space, and $R_t(p)$ penalizes candidate paths passing near geometrically supported occupied regions in the imagined scenes. The weights α_{sem}, α_{exp}, and α_{risk} balance target seeking, exploration, and safety.

Specifically, $S_t(p)$ uses exponentially decayed nearest distances to observed and imagined target-labeled Gaussians, where imagined target cues are discounted by a factor $\lambda_{\text{img}} < 1$ to avoid over-trusting generated content. $E_t(p)$ estimates the amount of previously unobserved but likely navigable space that can be estimated around p, computed from imagined free-space primitives not yet covered by the observed map. $R_t(p)$ is a soft geometric risk penalty computed from filtered and spatially deduplicated imagined occupied voxels according to their distances from the candidate path. All three component scores are normalized to $[0, 1]$ before fusion, and the combined FAVM score is normalized once more. Implementation details, including kernel radius, semantic voting, and normalization, are provided in Appendix D.

Fusion with observation-based cues. FAVM provides foresight, but navigation must remain grounded in current observations. We combine FAVM with an observation-based multi-sourced value map $m_t = m_t^{\text{act}} + m_t^{\text{sem}} + m_t^{\text{traj}} + m_t^{\text{heu}}$, which aggregates action guidance, semantic landmark cues, trajectory constraints, and heuristic navigation priors.

The final affordance map is computed as

$$m_t^{\text{aff}}(p) = \beta\, \bar{m}_t(p) + (1 - \beta)\, m_t^{\text{FA}}(p), \qquad p \in \mathcal{P}_t,$$

where $\bar{m}_t$ is the normalized observation-based value map and $\beta \in [0, 1]$ controls the balance between grounded evidence and imagined future guidance. The navigation target is selected as

$p_t^\star = \arg\max_{p \in \mathcal{P}_t} m_t^{\text{aff}}(p)$, with walkability constraints enforced during selection. Detailed definitions of FAVM scores and observation-based component maps are provided in Appendix D.

Execution. Following an instruction-driven hierarchical planner, the goal instruction is parsed into high-level action-landmark guidance, which contributes to the observation-based value map. The selected navigation target $p_t^\star$ is passed to the local planner to compute executable waypoints for robot control. To reduce latency, the world model is invoked periodically rather than at every navigation step.

4 Experiments

We evaluate Schrödinger's Navigator in both simulation and real-world settings to assess its navigation performance, component effectiveness, and practical deployability. Our experiments first compare against strong ZSON baselines on HM3D, then ablate the key design choices behind future imagination and FAVM-based planning, and finally validate the system on a physical Go2 quadruped in indoor environments. More qualitative results and failure cases are shown in Appendix F.

4.1 Experimental Setup

Baselines and benchmark. We compare our approach against a broad set of representative baselines, including task-trained approaches (ZSON [9], PixNav [13], SPNet [48], SGM [35]), zero-shot semantic or map-based explorers (CoW [10], ESC [28], L3MVN [49], TriHelper [50], VoroNav [51], GAMap [33], BeliefMapNav [30]), imagination-based VLM navigators (ImagineNav [46], ImagineNav++ [47]), and LLM/VLM-based planners (VLFM [12], InstructNav [31]). We conduct large-scale simulation evaluation on HM3D [52] within Habitat [53]. Following prior work, we report Success Rate (SR), Success weighted by Path Length (SPL), and Distance to Goal (DTG). Detailed descriptions of baselines are provided in Appendix E.

Implementation. We use Qwen3.6-35B-A3B [54] for instruction parsing and navigation-direction judgment, FlashWorld [19] for trajectory-conditioned 3D imagination, and GLEE [55] for semantic segmentation. In HM3D simulation, FlashWorld is triggered every 50 observation updates, rather than at every step. Real-world tests use a Unitree Go2 with an RGB-D camera in Office, Classroom, and Common Room scenes, covering static-object search and occlusion-heavy unknown-risk cases. Hyperparameters, runtime, compute, and deployment details are provided in Appendices A and G.

4.2 Simulation Results

4.2.1 Quantitative Results

Table 1 presents the quantitative results on HM3D. Our method achieves the highest SR of 0.682, outperforming the strongest baseline, ImagineNav++ (0.625), by 5.7 percentage points. It also attains the lowest reported DTG of 1.84 m, compared with 2.89 m for InstructNav. Meanwhile, our method achieves the second-highest SPL of 0.311, remaining competitive in path efficiency. These results demonstrate the effectiveness of future-aware navigation in improving target discovery and goal proximity while maintaining competitive path efficiency. Further analysis across different occlusion severity levels is provided in Appendix E (Table 6).

4.2.2 Ablation Analysis

To assess the key design choices of Schrödinger's Navigator, we ablate the trajectory ensemble size N, the fusion coefficient β, and the FAVM scoring components. All variants change one factor from the full model, except for the no-imagination baseline.

The results in Table 2 show that future imagination is effective only when properly sampled and grounded. Our full model improves SR from 0.553 to 0.682 over the no-imagination baseline, a gain

Table 1: **Quantitative Comparison on Simulation Results.** Cell background colors indicate the method is the best, second best, or third best on this metric.

Method	Training Free	HM3D		
		SR↑	SPL↑	DTG↓
ZSON [9]	✗	0.255	0.126	–
PixNav [13]	✗	0.379	0.205	–
SPNet [48]	✗	0.312	0.101	–
SGM [35]	✗	0.602	0.308	–
ESC [28]	✓	0.392	0.223	–
VLFM [12]	✓	0.525	0.304	–
VoroNav [51]	✓	0.420	0.260	–
L3MVN [49]	✓	0.504	0.231	4.43
TriHelper [50]	✓	0.565	0.253	3.87
GAMap [33]	✓	0.531	0.260	–
InstructNav [31]	✓	0.510	0.187	2.89
ImagineNav [46]	✓	0.530	0.238	–
ImagineNav++ [47]	✓	0.625	0.328	–
BeliefMapNav [30]	✓	0.614	0.306	–
Ours	✓	0.682	0.311	1.84

Table 2: Ablation and runtime analysis of Schrödinger's Navigator on HM3D. We report success rate (SR), average number of imagination triggers per episode, per-trigger imagination latency, episode-level waiting ratio, and waypoint change rate.

Variant	SR ↑	Trig. / Ep. ↓	Lat. / Trig. (s) ↓	Wait Ratio (%) ↓	Waypoint Change (%)
No Imagination	0.553	0.00	–	0.0	–
Single Traj. ($N = 1$)	0.594	4.31	3.0	7.2	28.4
Dense Sampling ($N = 5$)	0.622	4.18	15.0	30.8	51.6
Imagination-heavy ($\beta = 0.2$)	0.497	4.36	9.0	22.9	67.2
Observation-heavy ($\beta = 0.8$)	0.595	4.28	9.0	19.9	35.3
FAVM w/o Exploration $E(p)$	0.629	4.20	9.0	19.8	40.8
FAVM w/o Semantic $S(p)$	0.605	4.27	9.0	20.5	38.4
FAVM w/o Risk $R(p)$	0.634	4.19	9.0	19.3	42.1
Ours ($N = 3, \beta = 0.6$)	**0.682**	4.24	9.0	20.1	46.8

of 12.9 percentage points. However, more trajectories do not necessarily yield better performance: $N = 1$ provides limited coverage, while $N = 5$ incurs higher latency and still underperforms our $N = 3$ design, demonstrating the coverage–cost advantage of tri-trajectory sampling.

The fusion and component ablations further show that imagined guidance must be calibrated by real observations. The imagination-heavy setting ($\beta = 0.2$) achieves an SR of 0.497, while the observation-heavy setting ($\beta = 0.8$) reaches 0.595. Both underperform the balanced setting ($\beta = 0.6$), highlighting the importance of balancing imagined future guidance with current observations. Removing any FAVM component reduces SR, with the largest drop caused by removing the semantic score $S(p)$.

The runtime results indicate that 3D future imagination introduces sparse rather than per-step overhead. Our full model triggers imagination 4.24 times per episode, with 9.0 seconds per trigger and a 20.1% waiting-time ratio. Dense sampling increases the waiting-time ratio to 30.8% without improving SR, showing that additional futures introduce extra cost without proportional benefit.

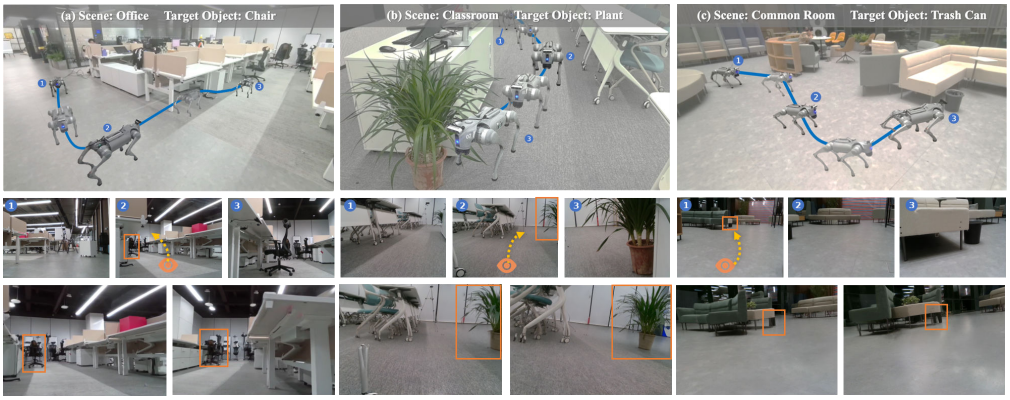


Figure 5: Real-world object-navigation demonstrations of Schrödinger's Navigator across three indoor scenarios: Office-Chair, Classroom-Plant, and Common Room-Trash Can. Each example shows the executed trajectory, egocentric observations with predicted target directions and target boxes, and world-model imagined views that reveal the target appearance.

Table 3: Real-world quantitative comparison against the InstructNav baseline. We report success counts (Success / Total Trials) across three indoor environments.

Scene	Office	Classroom	Common Room	All
Search for static objects				
InstructNav	7/10	7/10	**8/10**	22/30
Ours	**8/10**	**8/10**	7/10	**23/30**
Scenarios with Unknown Risks				
InstructNav	4/10	3/10	4/10	11/30
Ours	**5/10**	**6/10**	**6/10**	**17/30**

The 46.8% waypoint change rate further indicates that imagined futures actively influence waypoint selection beyond the observation-only map.

4.3 Real-World Deployment

Our real-world experiments provide a controlled evaluation against an InstructNav-style baseline, complementing the broader comparisons with navigation methods conducted on HM3D.

We evaluate adaptability under two task settings: (1) **Static Object Search**, where the agent explores standard scenes to locate objects; and (2) **Unknown-Risk Scenarios**, where severe static occlusions are not visible from afar, testing the agent's ability to anticipate uncertainty and avoid blind spots.

4.3.1 Performance Analysis

As shown in Table 3, we report success counts over 30 trials per setting, with 10 trials in each of the three environments.

In standard static object search, Schrödinger's Navigator performs comparably to InstructNav, achieving a slightly higher success rate ($23/30$ vs. $22/30$), indicating that future-aware reasoning does not compromise regular object-search performance. In scenarios with unknown risks, our method shows a clearer advantage, succeeding in $17/30$ trials compared with InstructNav's $11/30$. This suggests that imagined 3D futures help the robot better handle occlusions, ambiguous observations, and potential blind spots.

Overall, these real-world results show that our method maintains strong static-search performance while improving robustness in risk-aware navigation. Qualitative examples are provided in Figure 5.

5 Conclusion

Schrödinger's Navigator addresses ZSON limitations in occluded environments by leveraging a 3D trajectory-conditioned world model to imagine and fuse plausible 3D futures into a unified scene. This training-free framework constructs a single affordance map encoding semantics, exploration, and risk. Real-world experiments on a Unitree Go2 demonstrate its robustness, achieving comparable performance in static tasks while outperforming the baseline in occlusion-heavy scenarios.

6 Limitations

Schrödinger's Navigator uses imagined 3D futures as plausible hypotheses rather than exact reconstructions of hidden space. While FAVM discounts generated cues and grounds decisions in current observations, inaccurate generation, semantic transfer errors, or imperfect depth alignment may still bias waypoint selection when imagined evidence is over-weighted. Our method performs risk-aware waypoint selection but does not provide formal safety guarantees. Future work could improve reliability through uncertainty-calibrated imagination, stronger multi-view consistency, and active verification of uncertain regions. In addition, the current tri-trajectory sampler focuses on the dominant forward occluder for efficiency, but more complex layouts with multiple occluders or non-frontal blind spots may require adaptive trajectory budgeting. Broader evaluation across more diverse environments, dynamic obstacles, sensors, and robot platforms, as well as integration with certified local planners, would further strengthen real-world deployment.

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Supplementary Video

The supplementary video shows occlusion-heavy HM3D episodes and Go2 demonstrations in the Office, Classroom, and Common Room scenes, including executed trajectories, egocentric observations, target predictions, and trajectory-conditioned imagined views.

A Hyperparameters, Runtime, and System Configuration

Unless otherwise stated, the same parameters are used in HM3D simulation and real-world Go2 deployment. All value maps are normalized to $[0, 1]$, and distances are measured in meters.

A.1 Trajectory-imagination hyperparameters

The sampler uses three virtual camera trajectories, $\mathcal{V} = \{L, R, U\}$, corresponding to left-bypass, right-bypass, and over-the-top futures. All three trajectories are used for future imagination rather than robot execution.

Table 4: Geometric parameters for adaptive occluder-aware virtual-camera trajectory sampling.

Parameter	Symbol	Value	Description
Trajectory ensemble size	N	3	Left, right, and over-the-top futures.
Camera poses per trajectory	N_{pose}	16	Initial camera poses before interpolation.
Nominal radial offset	δ	0.20 m	Offset used to define the nominal orbit radius.
Minimum orbit radius	$d_c^{\min}$	1.0 m	Lower bound on the nominal orbit radius.
Nominal orbit radius	d_c	–	$d_c = \max(r_t + r_{\text{robot}} + \delta, d_c^{\min})$.
Angular expansion	$\Delta\theta$	20°	Extends lateral views beyond the observed occluder boundary.
Nominal minimum arc extent	$d_v^{\min}$	1.0 m	Lower setting for the nominal arc extent; realized trajectories may be shorter.
Maximum trajectory length	$d_v^{\max}$	3.0 m	Upper bound on the sampled camera path length.
Lateral edge-gaze weight	γ_{edge}	0.35	Interpolates the lateral gaze target toward the corresponding boundary side.

Table 5: Default coefficients for FAVM construction and fusion.

Parameter	Symbol	Value	Description
Observation/future fusion balance	β	0.60	Balances $\bar{m}_t$ and m_t^{FA}.
Semantic score weight	α_{sem}	1.0	Weight for target relevance $S_t(p)$.
Exploration score weight	α_{exp}	1.0	Weight for expected reveal $E_t(p)$.
Risk score weight	α_{risk}	1.0	Weight for traversal risk $R_t(p)$.
Imagined semantic discount	λ_{img}	0.50	Discount for imagined target cues.
Semantic-transfer depth tolerance	$\tau_{d,i}^{(m)}$	Adaptive	Radius- and depth-dependent; see Appendix C.

A.2 FAVM, perception, and runtime settings

We use Qwen3.6-35B-A3B [54] for instruction parsing and direction judgment, FlashWorld [19] for trajectory-conditioned 3D imagination, and GLEE [55] for open-vocabulary semantic segmentation. FlashWorld uses 480×704 input resolution, 24 conditioning camera poses interpolated from the 16 initial poses, and sparse triggering. Semantic transfer uses three rendered views; optional videos are exported at 15 fps.

A.3 Runtime characteristics

Trig./Ep. denotes FlashWorld invocations per episode; **Lat./Trig.** denotes per-trigger world-model service time; **Wait Ratio** is the fraction of episode time waiting for fresh future evidence; and **Waypoint Change** measures the fraction of navigation decisions in which the selected waypoint differs from that selected using the observation-only map. In HM3D simulation, FlashWorld is triggered at a fixed interval of 50 observation updates. In real-world execution it runs asynchronously while the local planner continues executing the current waypoint. In our experiments, $N = 1/3/5$ futures take about $3.0/9.0/15.0$s per trigger. The full model triggers 4.24 times per episode, with a 20.1% waiting ratio and a 46.8% waypoint-change rate.

B Occluder-Aware Trajectory Sampling

The sampler generates a compact set of virtual camera trajectories around the dominant forward occluder rather than densely enumerating actions. It is designed to cover complementary lateral and vertical reveal directions.

B.1 Point cloud preprocessing and occluder selection

Given RGB-D observation (I_t, D_t), intrinsics K, and pose q_t, valid depth pixels are back-projected as

$$x(p) = D_t(p)K^{-1}\bar{p}, \qquad \bar{p} = [u, v, 1]^\top. \tag{1}$$

Ground points are removed, and the remaining non-ground points are clustered into $\mathcal{C}_t = \{C_1, \ldots, C_M\}$. A cluster is a forward candidate if

$$C_j \in \mathcal{C}_t^{\mathrm{fwd}} \iff |\mathrm{wrap}\left(\angle(\bar{x}_j - x_t) - \psi_t\right)| \le \theta_{\mathrm{fwd}}. \tag{2}$$

We rank each forward cluster by normalized projected area, inverse distance, and heading alignment:

$$\rho_j = \overline{A}_j + (1 - \overline{d}_j) + \overline{h}_j, \qquad h_j = \max\{0, \cos(\angle(\bar{x}_j - x_t) - \psi_t)\}, \tag{3}$$

and select $O_t = \arg\max_{C_j \in \mathcal{C}_t^{\mathrm{fwd}}} \rho_j$.

B.2 Boundary, orbit center, and angular span

The selected occluder O_t is projected onto the ground plane, and its boundary B_t is extracted using a convex hull [56] or an alpha-shape contour [57]. We define

$$c_t = \frac{1}{|B_t|} \sum_{b \in B_t} b, \qquad r_t = \max_{b \in B_t} \|b - c_t\|_2, \qquad d_c = \max(r_t + r_{\mathrm{robot}} + \delta, d_c^{\min}). \tag{4}$$

For each boundary point b, let $\theta(b) = \operatorname{atan2}(b_y - c_{t,y}, b_x - c_{t,x})$. After unwrapping, the occluder span is $[\theta_{\min}, \theta_{\max}]$. Lateral target angles and arc lengths are

$$\theta_L = \theta_{\max} + \Delta\theta, \quad \theta_R = \theta_{\min} - \Delta\theta, \quad d_v^{(v)} = \operatorname{clamp}\big(d_c |\operatorname{wrap}(\theta_v - \theta_0)|, d_v^{\min}, d_v^{\max}\big), \tag{5}$$

where $v \in \{L, R\}$ and θ_0 is the current camera azimuth around c_t. The clamped length specifies a nominal arc extent; the realized start-anchored camera trajectory may be shorter than $d_v^{\min}$.

B.3 Trajectory instantiation and filtering

Based on the occluder geometry, the sampler generates continuous left, right, and overhead virtual camera trajectories from the current camera pose. Each trajectory is represented as

$$\tau_t^{(v)} = \{(p_k^{(v)}, R_k^{(v)})\}_{k=0}^{N_{\text{pose}}-1}, \quad v \in \{L, R, U\}. \tag{6}$$

The lateral trajectories follow occluder-aware arcs with angular extents determined by the observed occluder boundary.

For each lateral trajectory, the horizontal camera gaze target is defined as

$$g_{t,xy}^{(v)} = (1 - \gamma_{\text{edge}}) c_{t,xy} + \gamma_{\text{edge}} e_{t,xy}^{(v)}, \quad v \in \{L, R\}, \tag{7}$$

where $e_{t,xy}^{(v)}$ denotes the corresponding observed boundary-side gaze target and $\gamma_{\text{edge}} = 0.35$. Camera orientations are smoothly interpolated toward the target viewing directions using spherical linear interpolation (SLERP).

For the overhead trajectory, the camera faces the central direction of the observed occluder boundary, while its height varies smoothly as

$$z_k^{(U)} = z_{\text{cam}} + h_{\max} \sin\left(\frac{\pi k}{N_{\text{pose}} - 1}\right). \tag{8}$$

The sampler validates the generated camera trajectories for pose validity, continuity, and compatibility with the world-model input. If no reliable forward occluder is identified or the adaptive trajectories fail to meet the generation requirements, the sampler falls back to the predefined fixed left, right, and overhead virtual camera trajectories.

C 3DGS Alignment, Semantic Label Transfer, and Scene Fusion

This section summarizes how trajectory-conditioned 3DGS futures are converted into a planning representation.

C.1 Representation and coordinate alignment

Each Gaussian is stored compactly as

$$g_i = [x_i, y_i, z_i, c_i^r, c_i^g, c_i^b, \text{rad}_i, \text{opa}_i, \ell_i], \tag{9}$$

where (x_i, y_i, z_i) is the center, (c_i^r, c_i^g, c_i^b) is color, rad_i and opa_i are simplified size and opacity, and ℓ_i is the semantic label. Before generation, the current pose and sampled trajectory are transformed into a local camera-centered frame $\hat{W}$:

$$\hat{q}_t = T_{\hat{W} \leftarrow W}(q_t), \qquad \hat{\tau}_t^{(v)} = T_{\hat{W} \leftarrow W} \tau_t^{(v)}, \qquad \hat{G}_t^{(v)} = \Phi_\theta(I_t, \hat{q}_t, \hat{\tau}_t^{(v)}). \tag{10}$$

After scale alignment, Gaussian centers are transformed back to the world frame:

$$\tilde{\mu}_i^{(v)} = R_t(s_t^{(v)} \hat{\mu}_i^{(v)}) + t_t. \tag{11}$$

The compact imagined primitives use a fixed radius of $0.03\,\text{m}$ and opacity of 1 after filtering.

C.2 Depth scale alignment and semantic transfer

For trajectory v, we render $\hat{D}_t^{(v)}$ from $\hat{G}_t^{(v)}$ at the current viewpoint and estimate scale by the median depth ratio:

$$\Omega_t^{(v)} = \{p \mid 0.1 < D_t(p) < 5.0, \hat{D}_t^{(v)}(p) > 10^{-4}\}, \qquad s_t^{(v)} = \mathrm{median}_{p \in \Omega_t^{(v)}} \frac{D_t(p)}{\hat{D}_t^{(v)}(p)}. \tag{12}$$

The observed depth is center-cropped and resized using nearest-neighbor interpolation before alignment. At least 100 valid pixels are required; otherwise, alignment raises an error. For semantic transfer, each available view m has a 2D semantic map $Y_t^{(m)}(p)$. A projected primitive receives a valid vote only if it is inside the image and depth-consistent:

$$\begin{aligned} p_i^{(m)} &= \pi(K T_{C_m \leftarrow W} \mu_i), \\ \left| z_i^{(m)} - D_{\mathrm{rend}}^{(m)}(p_i^{(m)}) \right| &\le \tau_{d,i}^{(m)}, \\ \tau_{d,i}^{(m)} &= \max\left\{ 2\,\mathrm{rad}_i,\ \frac{D_{\mathrm{rend}}^{(m)}(p_i^{(m)})}{f_{\mathrm{pix}}^{(m)}} \right\}. \end{aligned} \tag{13}$$

Here $z_i^{(m)}$ is the positive camera-frame depth of the Gaussian center, and $f_{\mathrm{pix}}^{(m)}$ is the smaller of the horizontal and vertical focal lengths in pixels. Only finite, positive rendered depths are used. Its final label is obtained by majority voting over valid semantic labels:

$$\ell_i = \arg\max_c \sum_{m \in \mathcal{M}_i} \mathbf{1}[Y_t^{(m)}(p_i^{(m)}) = c]. \tag{14}$$

Segmentation confidence is thresholded at 0.3 before voting; primitives with no valid votes remain unlabeled.

C.3 Scene fusion

The aligned imagined primitives are first collected as

$$G_t^{\mathrm{img}} = \bigcup_{v \in \mathcal{V}} \tilde{G}_t^{(v)}, \qquad G_t^{\mathrm{fuse}} = \mathrm{Merge}(G_t^{\mathrm{obs}} \cup G_t^{\mathrm{img}}). \tag{15}$$

Here Merge denotes the joint use of separately stored observed and imagined representations, rather than confidence-weighted parameter averaging. The source indicator $\sigma_g \in \{\mathrm{obs}, \mathrm{img}\}$ is determined by buffer membership. Imagined trajectory outputs are concatenated after per-trajectory filtering and voxel downsampling, while the observed map is accumulated and downsampled separately. Voxel downsampling retains one representative primitive per occupied voxel.

D Future-Aware Value Map and Observation-Based Component Maps

FAVM is evaluated on candidate waypoints using G_t^{fuse}, while σ_g separates observed evidence from imagined hypotheses.

D.1 FAVM components

Let $\mathcal{P}_t = \{\mu_g \mid g \in \mathcal{G}_{\mathrm{nav}}\}$ denote candidate navigation positions given by the centers of navigable Gaussians. All scalar maps are normalized over $\mathcal{P}_t$ by

$$\mathcal{N}(x)(p) = \frac{x(p) - \min_{q \in \mathcal{P}_t} x(q)}{\max_{q \in \mathcal{P}_t} x(q) - \min_{q \in \mathcal{P}_t} x(q) + \epsilon}. \tag{16}$$

For target category $y^\star$, observed and imagined target sets are

$$T_t^{\mathrm{real}} = \{g \in G_t^{\mathrm{fuse}} \mid \ell_g = y^\star, \sigma_g = \mathrm{obs}\}, \quad T_t^{\mathrm{hyp}} = \{g \in G_t^{\mathrm{fuse}} \mid \ell_g = y^\star, \sigma_g = \mathrm{img}\}. \tag{17}$$

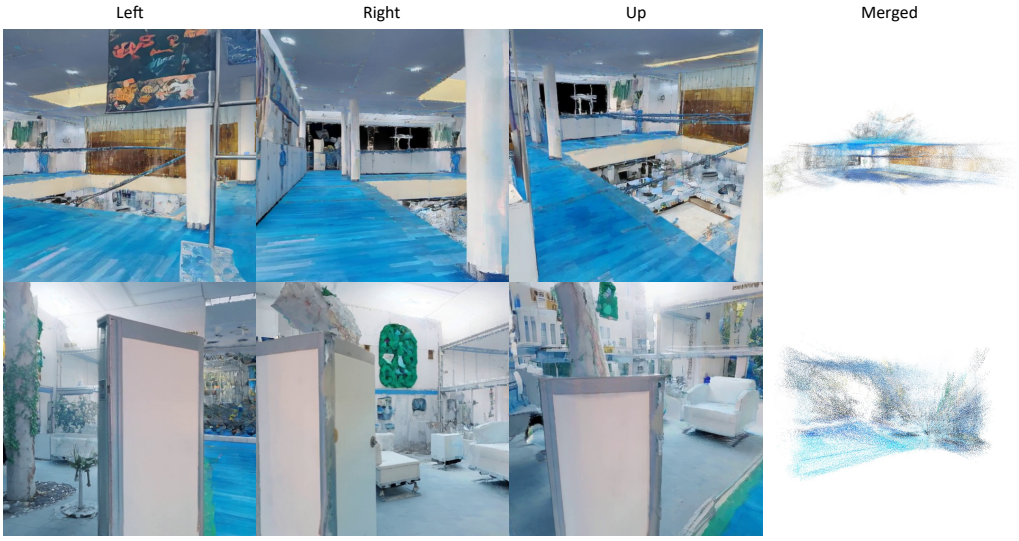


Figure 6: Trajectory-conditioned 3DGS generation and fusion. Each row shows left-bypass, right-bypass, over-the-top, and merged futures.

The semantic score discounts imagined target cues:

$$\tilde{S}_t(p) = e^{-d_t^{\text{real}}(p)/\tau_{\text{sem}}} + \lambda_{\text{img}} e^{-d_t^{\text{hyp}}(p)/\tau_{\text{sem}}}, \quad S_t = \mathcal{N}(\tilde{S}_t). \tag{18}$$

Here $d_t^a(p) = \min_{g \in T_t^a} \|p - \mu_g\|_2$ for $a \in \{\text{real}, \text{hyp}\}$, $\tau_{\text{sem}} = 1.0\,\text{m}$, and $\lambda_{\text{img}} = 0.5$. An empty target set contributes zero. At each imagination update, F_t^{new} is constructed from imagined primitives lying less than $0.20\,\text{m}$ from the estimated floor height and more than $0.15\,\text{m}$ from the observed navigable map, followed by voxel downsampling. The expected reveal score is approximated by local coverage:

$$\tilde{E}_t(p) = \sum_{g \in F_t^{\text{new}}} \mathbf{1}[\|p - \mu_g\|_2 \le r_{\text{vis}}], \quad E_t = \mathcal{N}(\tilde{E}_t). \tag{19}$$

Here $r_{\text{vis}} = 0.60\,\text{m}$. For risk estimation, we extract a set of imagined occupied voxels H_t^{img} through depth-consistency checks, geometric filtering, and spatial deduplication. Let $d_{\text{XY}}(h, p)$ denote the shortest ground-plane distance from voxel h to the line segment connecting the current camera position and candidate waypoint p. The geometric risk score is computed as

$$\tilde{R}_t(p) = \sum_{h \in H_t^{\text{img}}} \exp\left(-\frac{d_{\text{XY}}(h,p)^2}{2r_{\text{path}}^2}\right), \qquad R_t = \mathcal{N}(\tilde{R}_t). \tag{20}$$

We set $r_{\text{path}} = 0.35\,\text{m}$. When no valid risk evidence is available, R_t is set to zero. The Future-Aware Value Map is

$$m_t^{\text{FA}}(p) = \mathcal{N}\left(\alpha_{\text{sem}} S_t(p) + \alpha_{\text{exp}} E_t(p) - \alpha_{\text{risk}} R_t(p)\right), \qquad p \in \mathcal{P}_t. \tag{21}$$

D.2 Observation-based map and navigation target

The observation-based map is

$$\bar{m}_t(p) = \mathcal{N}\left(m_t^{\text{act}}(p) + m_t^{\text{sem}}(p) + m_t^{\text{traj}}(p) + m_t^{\text{heu}}(p)\right). \tag{22}$$

Here m_t^{sem} favors proximity to currently observed landmarks, m_t^{act} encodes the next high-level action sector or frontier preference, m_t^{traj} discourages revisits using trajectory history, and m_t^{heu} projects the multimodal planner's predicted direction onto the map. The final affordance map and navigation target are

$$m_t^{\text{aff}}(p) = \beta \bar{m}_t(p) + (1-\beta) m_t^{\text{FA}}(p), \qquad p_t^\star = \arg\max_{p \in \mathcal{P}_t} m_t^{\text{aff}}(p). \tag{23}$$

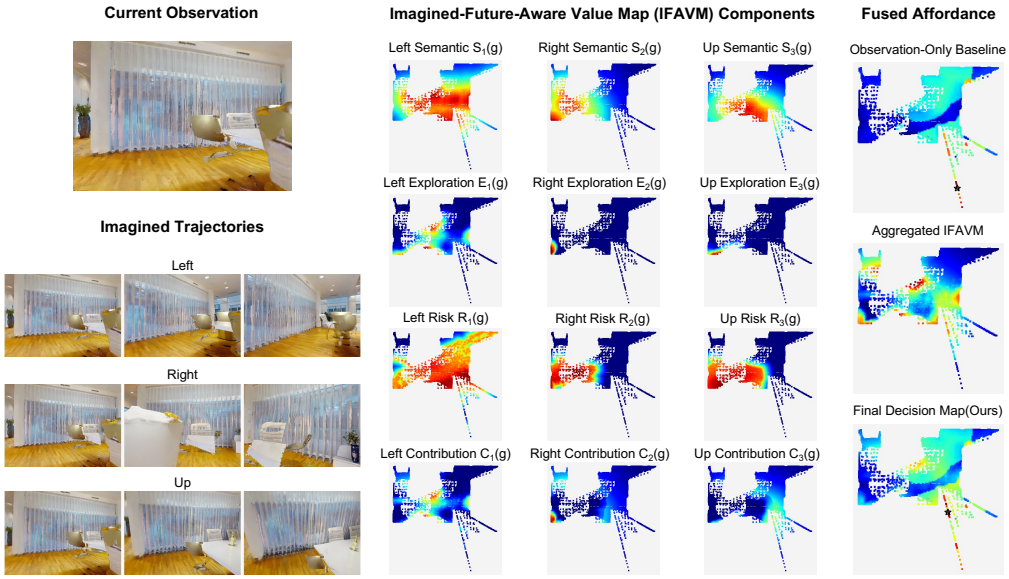


Figure 7: Future-Aware Value Map construction. The figure shows the current observation, left/right/up imagined futures, per-trajectory semantic, exploration, risk, and contribution maps, and the final fused affordance map. Warm colors indicate higher utility, except in risk maps where they indicate higher penalty; the in-figure label IFAVM denotes the FAVM decomposition.

The selected navigation target is used by the local planner to generate executable waypoints. Between triggers, the robot uses the latest observation-based map and the most recent valid FAVM.

For visualization, the per-trajectory contribution is

$$C_t^{(v)}(p) = \mathcal{N}\left(\alpha_{\text{sem}} S_t^{(v)}(p) + \alpha_{\text{exp}} E_t^{(v)}(p) - \alpha_{\text{risk}} R_t^{(v)}(p)\right). \tag{24}$$

Figure 7 shows that FAVM shifts value from visible frontiers toward regions that balance target relevance, expected reveal, and traversal safety.

E Baselines, Evaluation Protocol, and Metrics

E.1 Simulation benchmark and metrics

We evaluate on HM3D within Habitat. Following the ObjectNav protocol, each episode is capped at a maximum of 500 steps and is considered successful if the agent stops within 1.0 m of an instance of the target category. We report Success Rate (SR), Success weighted by Path Length (SPL), and Distance to Goal (DTG). SR measures the fraction of successful episodes. SPL rewards successful episodes with shorter paths. DTG measures the final geodesic distance from the agent to the target object. In HM3D simulation, the selected waypoints are executed using Habitat's `ShortestPathFollower` with discrete navigation actions.

We compare against four groups of baselines.

Task-trained approaches. ZSON [9], PixNav [13], SPNet [48], and SGM [35] rely on task-specific training or learned navigation policies. These methods can perform well in their training domains but are less aligned with the training-free zero-shot setting targeted by our method.

Zero-shot geometric and semantic explorers. ESC [28], L3MVN [49], TriHelper [50], VoroNav [51], GAMap [33], and BeliefMapNav [30] use geometric frontiers, semantic maps, language priors, or belief maps to guide exploration without task-specific fine-tuning. They improve semantic

Table 6: HM3D success rates stratified by occlusion severity. ΔSR denotes Ours minus ImagineNav++.

Occlusion	ImagineNav++ SR ↑	Ours SR ↑	ΔSR ↑
Low	0.682	0.720	+0.038
Medium	0.629	0.674	+0.045
High	0.564	0.652	+0.088

exploration but typically estimate value from observed frontiers or a single belief state rather than from an ensemble of trajectory-conditioned future 3D scenes.

LLM/VLM-based planners. VLFM [12] and InstructNav [31] use vision-language or language-model reasoning to construct navigation guidance. For real-world comparison, we use a Qwen3.6-35B-A3B-based implementation of InstructNav-style planning to align the high-level reasoning capability with our system. This baseline uses the same RGB-D observations and local controller where applicable, but it does not use trajectory-conditioned 3D future imagination or FAVM.

Imagination-based navigators. ImagineNav [46] and ImagineNav++ [47] use scene imagination to assist embodied navigation. Unlike these methods, Schrödinger's Navigator imagines 3DGS futures conditioned on multiple candidate trajectories and fuses the resulting semantic, exploratory, and risk evidence into a spatial FAVM for zero-shot ObjectNav.

As occlusion severity increases, both methods exhibit lower success rates, while the advantage of our method becomes more pronounced. Our method outperforms ImagineNav++ by 3.8, 4.5, and 8.8 percentage points under low, medium, and high occlusion, respectively.

E.2 Real-world protocol

Real-world experiments are conducted on a Unitree Go2 quadruped equipped with an RGB-D camera in three indoor environments: Office, Classroom, and Common Room. We evaluate two settings. In Static Object Search, the target is a static object in a standard indoor scene. In Unknown-Risk Scenarios, the scene includes severe occlusion or blind spots that may hide targets or risky regions from the initial viewpoint. Each scene contains 10 trials per setting, and success is reported as Success / Total Trials.

F Additional Qualitative Results and Failure Analysis

In qualitative overlays, **Flag** denotes the visual target-verification signal and **DTS** denotes the logger's distance-to-success value; main quantitative results use DTG.

Successful rollouts. Figure 8 shows cases where imagined futures guide the agent toward viewpoints that reveal hidden targets across bed, potted plant, toilet, chair, sofa, and television set episodes.

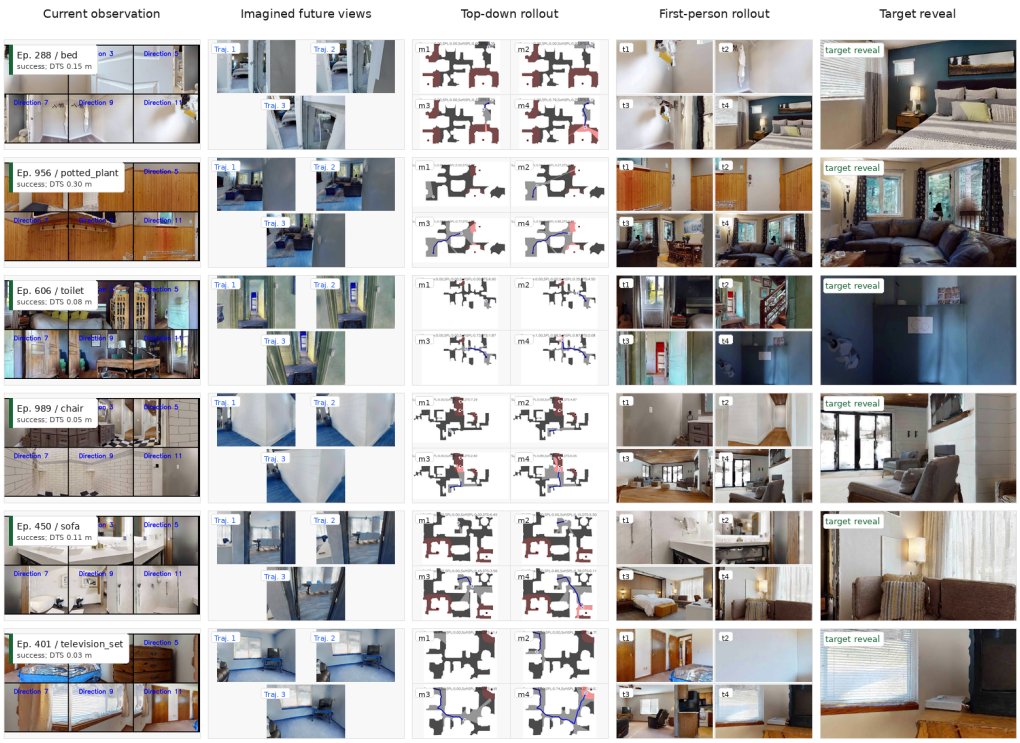


Figure 8: Successful occlusion-heavy simulation rollouts. Each row shows the current observation, imagined future views, top-down checkpoints, first-person rollout frames, and the final target reveal.

End-to-end failures. Figure 9 summarizes localization, detection, occlusion-handling, and termination failures, which typically arise from compounding perception, mapping, planning, and stopping errors.

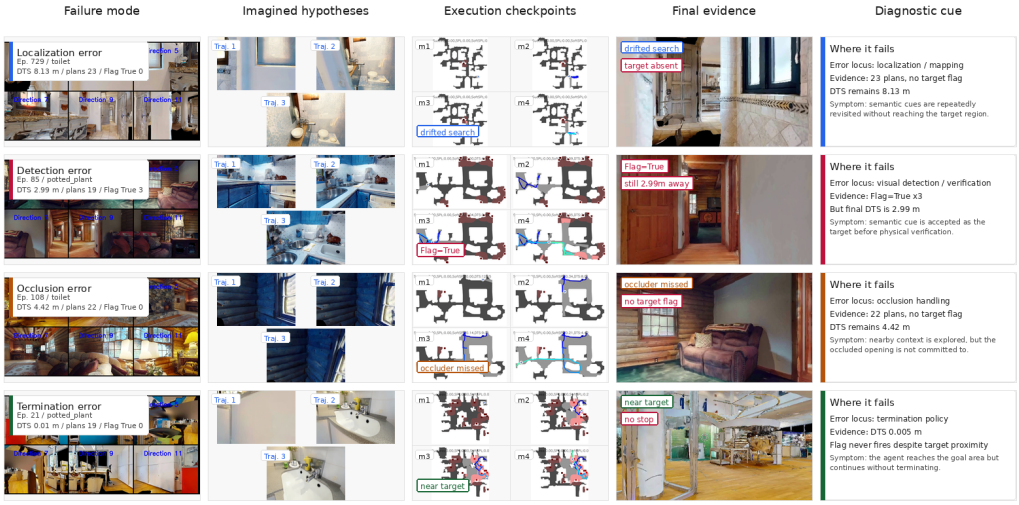


Figure 9: End-to-end failure modes. Rows show localization, detection, occlusion-handling, and termination errors with imagined hypotheses, execution checkpoints, final evidence, and diagnostic cues.

World-model failures. Figure 10 isolates hallucinated target evidence, scale/depth drift, missed occluded evidence, and trigger gaps. These cases motivate discounting imagined semantics, penalizing geometrically supported occupied regions along candidate paths, and fusing future evidence with observation-based maps.

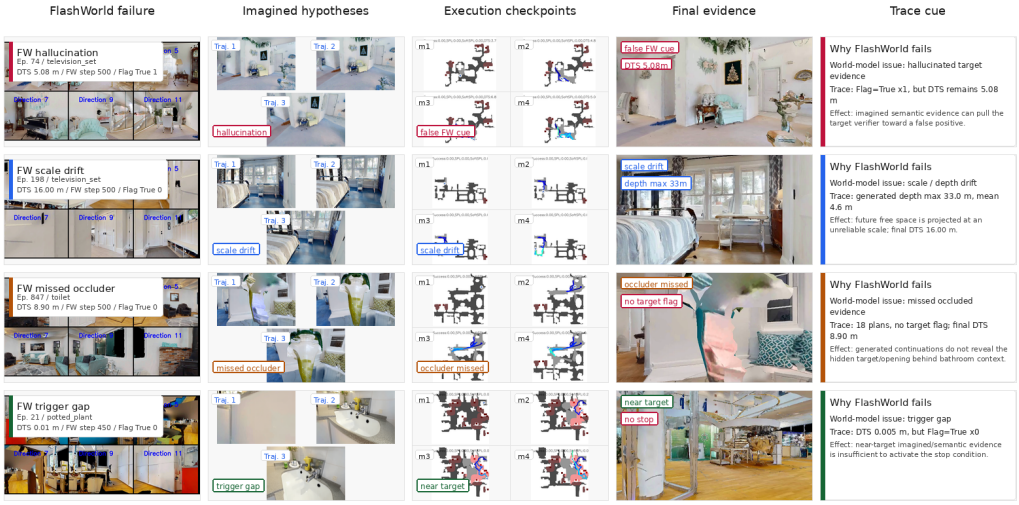


Figure 10: FlashWorld-related failure modes. Rows show hallucinated target evidence, scale/depth drift, missed occluded evidence, and trigger gaps.

G Real-World Deployment Details

G.1 Hardware, services, and safety

Real-world experiments use a Unitree Go2 quadruped with an Intel RealSense D435i RGB-D camera. The onboard Jetson Orin maintains robot-state feedback and velocity control, while RGB-D observations are forwarded through FastAPI to a remote GPU server for perception, future imagination, and FAVM waypoint selection. The asynchronous split keeps high-frequency control independent of the slower world-model service.

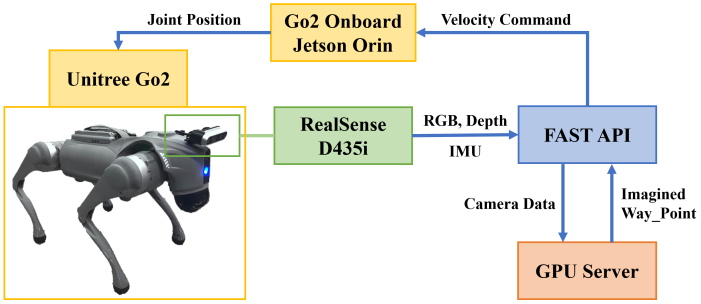


Figure 11: Real-world deployment architecture for Go2 sensing, FastAPI communication, remote GPU inference, and velocity-command execution.

For HM3D simulation, four RTX 4090 GPUs support parallel episode evaluation and shared model inference, increasing evaluation throughput without changing the per-episode navigation protocol.

Each control cycle updates the local point cloud and semantic/navigable maps, parses action-landmark guidance, triggers future imagination when needed, fuses observation-based maps with FAVM, and sends the selected waypoint to the local planner. In Static Object Search, targets are

Table 7: Compute and service configuration.

Item	Configuration
Robot platform	Unitree Go2 quadruped
Onboard computer	Go2 onboard Jetson Orin
RGB-D sensor	Intel RealSense D435i
Simulation compute	Four NVIDIA RTX 4090 GPUs (parallel evaluation)
Real-world inference GPU	One NVIDIA H800 GPU on a remote server
World-model service	FlashWorld on the remote GPU server via FastAPI
High-level reasoning	Qwen3.6-35B-A3B for instruction and direction judgment
Control interface	FastAPI waypoint service with velocity-command output

common indoor objects in plausible locations; in Unknown-Risk Scenarios, targets or risky regions are partially hidden by furniture, dividers, desks, or plants. The robot executes only locally collision-checked waypoints. Imagined futures modify affordance scores but never directly command motion, and all real-world trials use conservative speed limits with manual supervision.